RetiNova

Finding diabetic eye disease earlier, for people who cannot reach an eye doctor.

Smart India Hackathon 2026 · Problem SIH25131
Judge brief

01 · THE PROBLEM

Diabetes quietly blinds people.

It damages the tiny blood vessels at the back of the eye. There is no pain and no warning.

By the time sight blurs, the damage is already done.

02 · THE SCALE

101 million

people in India live with diabetes

1 in 3

of them will develop eye disease

1 : 100,000

eye doctors to people

03 · THE REAL BOTTLENECK

The knowledge exists. The queue is the problem.

Everyone with diabetes should have their eyes checked every year.

There are not enough specialists to do it, and most of them are in cities.

04 · WHAT WE BUILT

A screening tool that decides who a doctor should see first.

It reads a retinal photograph, gives a severity grade, and says whether a specialist is needed.

A doctor confirms every single one. It never decides alone.

05 · WHO USES IT

The patient or health worker

Opens a website on a phone. Takes a photo. Gets a written report.

The doctor

Sees a queue sorted by risk, checks the AI's reasoning, signs the decision.

06 · THE DATA

Where the model learned.

35,126 retinal photographs, already graded by human doctors.

huggingface.co/datasets/youssefedweqd/Diabetic_Retinopathy_Detection
MIT licence — free to use.

07 · THE DATA, SPLIT UP

PURPOSE	IMAGES
Learning from	25,290
Tuning with	2,810
Locked away for testing	7,026

The test images were never seen during training. Every number we quote comes from them.

08 · WHERE IT ORIGINALLY CAME FROM

EyePACS, through Kaggle.

A US retinal screening network. The images were used in a Kaggle competition in 2015.

kaggle.com/competitions/diabetic-retinopathy-detection

> Said honestly: the copy we used does not name its source. We matched it — same image count, same grading scale. It is a strong inference, not a documented fact.

09 · THE CATCH IN THE DATA

Most eyes are healthy.

GRADE	TRAINING IMAGES
0 — no disease	18,583
1 — mild	1,759
2 — moderate	3,811
3 — severe	628
4 — proliferative	509

Real screening data looks exactly like this. It is why we never judge the model on accuracy.

10 · THE HONESTY TEST

Testing on your own data flatters you.

So we ran the finished model on photographs from two other countries and two other cameras.

Nothing was retrained. Nothing was adjusted to make the numbers look better.

11 · TEST ONE

IDRiD — Indian patients.

516 photographs taken on a Kowa camera in Nanded, Maharashtra.

The closest public data to the people this project is built for.

12 · TEST TWO

Messidor-2 — French patients.

1,744 photographs on a Topcon camera.

The dataset most published research is measured against.

TESTED ON	DISEASE CAUGHT	HEALTHY CLEARED
Its own test set	80.9%	60.3%
Indian patients	95.4%	65.3%
French patients	82.9%	51.4%

Same model. Same settings. Three different populations.

14 · WHAT THAT PROVES

It learned the disease, not the camera.

On Indian eyes it found 308 of the 323 cases that needed a specialist.

A model that had merely memorised its training images would have collapsed here.

15 · WHERE IT IS WEAK

The middle grade.

It tells obviously healthy from obviously diseased well. Grade 2 — moderate — it often misplaces.

We know it. We measure it. It is why a doctor checks every case.

16 · THE HONEST NUMBER

1 in 5

cases needing a specialist are still missed

It is printed on every report we produce. A low grade never means "you are fine".

Three pieces.

PIECE	JOB
Website	Camera, upload, results, reports
Main server	Accounts, records, doctor workflow
AI server	Image checks and grading

18 · THE WEBSITE

React, running in the browser.

No app to install. The camera works from the web page itself.

React 19 · Vite · React Router

19 · THE MAIN SERVER

Spring Boot, in Java.

Logins, patient records, the doctor's queue, the review and the sign-off.

Spring Boot 4.1 · Java 17

20 · THE AI SERVER

Python, kept separate.

So the model can be retrained and replaced without touching patient records.

FastAPI · ONNX Runtime · OpenCV

21 · THE DATABASE

MongoDB.

One screening is one record: the image, the grade, the quality score, the doctor's decision.

The images live inside the record, so they cannot go missing when a server restarts.

- Passwords are scrambled, never stored.
- Signed tokens prove who you are on every request.
- Patients and doctors see different screens and different data.
- No one can open another patient's images.

23 · LIVE UPDATES

The doctor's queue updates by itself.

A new screening appears in under a second. Nobody presses refresh.

24 · THE MODEL

EfficientNetB0.

A neural network small enough to run on an ordinary computer, with no graphics card.

5.3 million settings, against 25 million or more for the usual alternatives.

16 megabytes. 13 milliseconds.

We convert the trained model into ONNX, a portable format.

The original needed 600 MB of software to run. This fits in a free server.

26 · SHOWING ITS REASONING

A heat map over the retina.

Red marks the parts of the image that pushed the grade.

The doctor checks a region instead of guessing what the machine was looking at.

27 · THE MOST IMPORTANT SAFETY FEATURE

It refuses bad photographs.

Show any AI a blurred picture and it will still give you a grade with a confident number next to it.

> A confident "no disease" on a photograph nobody could read is the most dangerous thing a screening tool can do. The person goes home reassured.

Nine quick tests before the AI runs.

IS THERE AN EYE?

- Is anything lit up
- Does it fill the frame
- Is it red, like a retina
- Does it have real texture

IS IT GOOD ENOUGH?

- Sharp enough
- Bright enough
- Not washed out
- Not glaring
- Roughly centred

POINTED AT	ANSWER
A healthy retina	Graded — quality 97 / 100
A diseased retina	Graded — referred
A wall	Nothing. Refused.
A hand over the lens	Nothing. Refused.
Random noise	Nothing. Refused.

Refusing takes 15 milliseconds. No score is shown at all.

30 · WHY THE CAMERA RUNS LIVE

To fix the photo while the eye is still there.

The screen says *move closer*, *too bright*, *hold still* — one instruction at a time.

The save button stays locked until the picture is actually usable.

31 · A SECOND PATH

What if there is no lens at all?

A phone camera cannot see the back of the eye. That needs a special lens.

So the plain camera measures the person instead — and still produces something useful.

32 · MEASUREMENT ONE

Pulse, from the skin.

Every heartbeat changes how much light the skin absorbs, by an amount too small to see.

Twenty seconds of video makes it clear.

33 · MEASUREMENT TWO

The pupil's reaction to light.

Shine a light, watch how fast it shrinks.

A slow or lopsided reaction is a recognised nerve warning sign in long-standing diabetes.

34 · MEASUREMENT THREE

Paleness of the inner eyelid.

A screening signal for anaemia, which is common in diabetic kidney disease.

The coloured part of the eye is used as a ruler, so distance and room light do not skew it.

35 · WHAT THAT PATH GIVES

An order, not a diagnosis.

Out of forty people waiting in a camp, who should be seen first.

No hardware. No lens. Works today on any phone.

We do not report the most likely grade.

We ask one question: is a specialist needed?

That is a single line drawn across the probability of grade 2 or worse.

METHOD	DISEASE CAUGHT	HEALTHY CLEARED
Most likely grade	49.7%	89.1%
A chosen line	80.9%	60.3%

The first looks better. It also sends half the sick people home.

38 · THE TRADE, STATED

We refer some healthy people on purpose.

An unnecessary appointment costs an hour. A missed case costs eyesight.

The report says this in plain words, every time.

39 · THE DOCTOR

Nothing is final without a human.

Every screening waits, marked "awaiting review", until a doctor signs it.

They see the image, the heat map, the grade and the quality score together.

Something the patient can keep.

- Their name, the date, which eye
- The grade, and what it means
- The images and the heat map
- What usually happens next
- What this test cannot tell them
- The doctor who signed it

41 · START TO FINISH

```
photo taken or uploaded
         │
         ▼
  is there an eye?  ──── no ──► say so, show nothing
         │ yes
         ▼
  is it good enough? ─── no ──► one instruction, try again
         │ yes
         ▼
  grade it, draw the heat map
         │
         ▼
  specialist needed?  yes / no
         │
         ▼
  doctor's queue  ──►  reviewed and signed
         │
         ▼
  patient downloads the report
```

42 · QUESTION

"Your accuracy is only 44%."

Accuracy is the wrong measure, and we do not use it.

Three quarters of these images are healthy eyes. A model that answered "healthy" every time would score 73% and be worthless.

43 · QUESTION

"Why is your specificity low?"

Because we chose it. Catching more disease always means more false alarms.

Moving that line is a single setting, and it is a decision for a health programme, not for us.

44 · QUESTION

"Does it work anywhere else?"

Yes — measured, not claimed.

Two other countries, two other cameras, nothing retrained. It caught between 81% and 95% of cases.

45 · QUESTION

"Can it run offline?"

Not today, and we will not claim it.

Nothing in the design prevents it — the model is 16 MB and needs no graphics card — but we have not built and tested it.

46 · QUESTION

"What is actually new?"

- It refuses pictures nobody could grade, before grading them.
- It works with no eye equipment at all, by measuring the person.
- It decides referral at a stated, chosen operating point.

47 · THE DISEASE

What diabetic retinopathy is.

Sugar damages the small vessels at the back of the eye. They leak, then close, then the eye grows fragile new ones that bleed.

Treatment stops it getting worse. It rarely gives back what is lost. That is why early matters.

SCREENING

Sorting many people into "look closer" and "not now". Allowed to over-refer.

We do the first. We never claim the second.

DIAGNOSIS

The closer look, by a doctor. Must be right.

49 · WHAT WE DO NOT CHECK

Macular oedema.

Swelling at the centre of the retina. It can blind someone at any grade.

Our images are not labelled for it, so we do not look for it — and the report says so.

STANDARD	COVERS
IEC 62304	How medical software must be built
ISO 14971	Managing risk in medical devices
ISO 27001	Information security
DPDP Act 2023	India's data protection law
CDSCO	India's medical device regulator

51 · WHERE WE HONESTLY STAND

A prototype, certified against none of them.

We know which rules would apply and what each one asks for.

Consent flows, audit logs and formal risk documents come before real patients. Saying otherwise would be the wrong answer.

TERM	MEANING
Sensitivity	Of sick people, how many we catch
Specificity	Of healthy people, how many we clear
Kappa	Agreement with human graders, chance removed
Grad-CAM	The heat map showing where it looked
ONNX	Portable format that made the model 16 MB

TERM	MEANING
DR	Diabetic retinopathy
ICDR	The international 0–4 grading scale
OD / OS	Right eye / left eye
PPG	Pulse read from skin colour change
PLR	The pupil's reaction to light
SaMD	Software treated as a medical device

54 · STILL TO DO

- Let the doctor correct the AI's grade, not just the decision.
- Compare a scan against the patient's previous ones.
- Build and test the offline version.
- Improve the middle grade.

55 · WHY THIS SHOULD WIN

Because it is measured, not claimed.

Every number here came from the running system.

Its failures are printed on the product itself, and it was tested on people it had never seen, in a country it was not trained on.

RetiNova

A screening aid. Not a diagnosis.

Not approved by any regulator. No result should be acted on without a doctor. Figures measured on public research datasets; clinical use would require prospective validation.